

Introduction to Supervised Learning

Applied Machine Learning — Session 1, Chapter 3

Supervised Learning

Learning from labeled examples to predict new cases.

Input: $\{ (X_1, y_1), (X_2, y_2), \dots, (X_n, y_n) \}$

↓

Model Training

↓

Output: $f(X_{\text{new}}) \rightarrow \hat{y}_{\text{new}}$

- **X** = features (what we know)
- **y** = label / target (what we want to predict)
- **$\hat{y}$** = prediction (what the model says)

Regression vs Classification

Task	Target	Example
Regression	Continuous number	House price: €285,000
Classification	Discrete category	Spam / Not Spam

The Learning Process

- ① Collect labeled data
- ② Choose a model
- ③ Train: minimize error between $\hat{y}$ and y
- ④ Evaluate on held-out data
- ⑤ Deploy or iterate

Training error = error on data the model has already seen

Test error = error on new, unseen data ← **what we actually care about**

Loss Functions

How we measure "how wrong" the model is:

Regression:

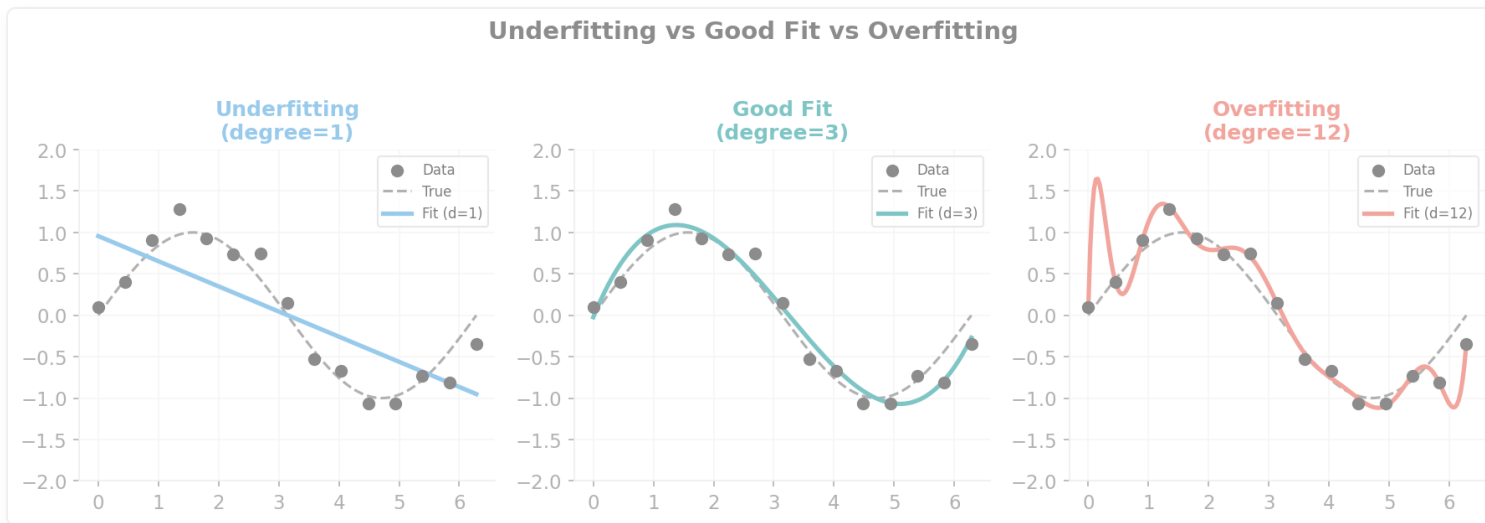
$$\text{MSE} = (1/n) \sum (y_i - \hat{y}_i)^2$$

Classification:

$$\text{Cross-Entropy Loss} = -\sum y_i \cdot \log(\hat{y}_i)$$

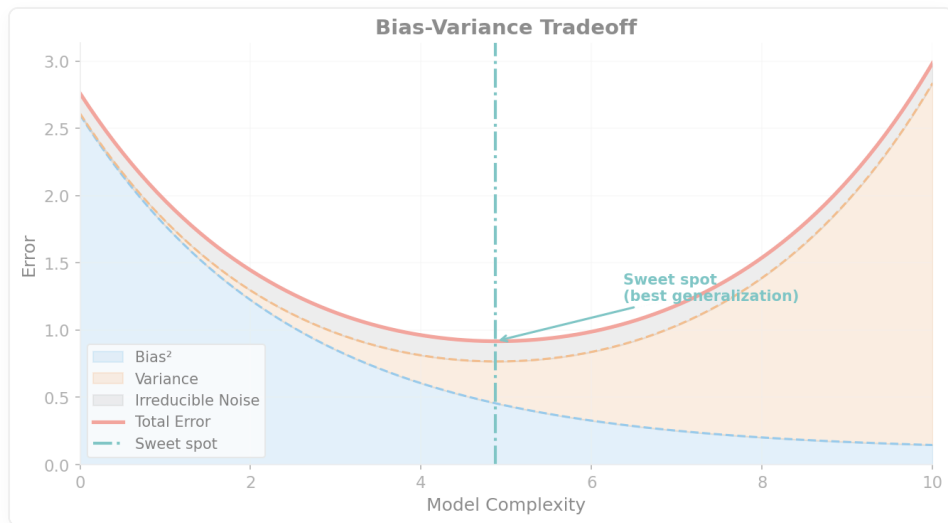
Training = finding parameters that **minimize** the loss.

Underfitting vs Overfitting



- **Underfitting:** High bias, high train error AND high test error
- **Overfitting:** Low bias, low train error BUT high test error

The Bias-Variance Tradeoff

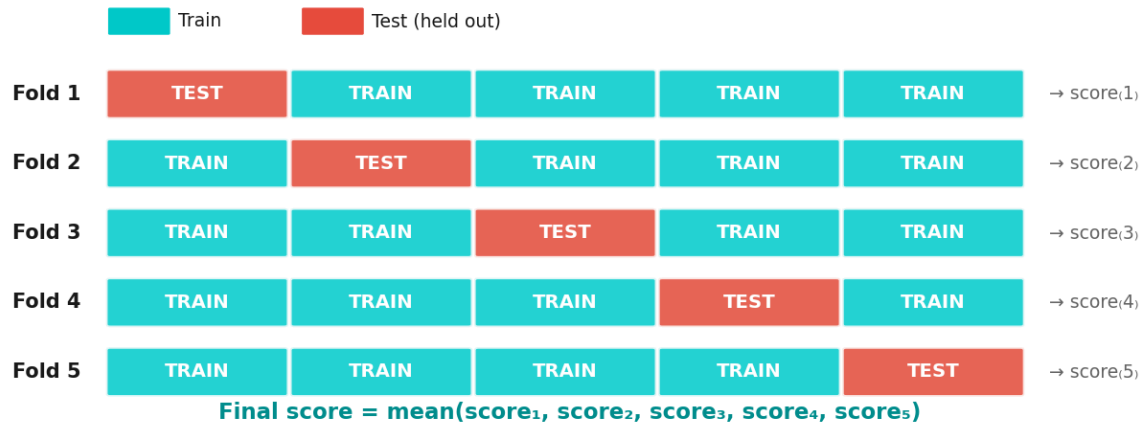


Goal: Find the sweet spot where total error is minimized.

Cross-Validation

Problem: Single train/test split gives noisy performance estimates.

5-Fold Cross-Validation



```
from sklearn.model_selection import cross_val_score
scores = cross_val_score(model, X, y, cv=5)
```

The sklearn API

One interface to rule them all:

```
# 1. Create
model = SomeModel(hyperparameter=value)

# 2. Train
model.fit(X_train, y_train)

# 3. Predict
y_pred = model.predict(X_test)

# 4. Score
model.score(X_test, y_test)
```

Every sklearn model works exactly this way. 

Now: Examples!

→ Open `02-examples/ch03_supervised_intro_examples.ipynb`

We'll see overfitting and underfitting live,
then fix it with cross-validation.

Key Takeaways

- Supervised learning = learning from (X, y) pairs
- Goal = generalize to **new, unseen data** (not memorize training data)
- Underfitting: model too simple
- Overfitting: model memorizes noise
- Cross-validation: robust performance estimation
- sklearn API: fit $\rightarrow$ predict $\rightarrow$ score

Next: Chapter 4

Regression Models

"Time to meet the algorithms. Starting with the classics."